Work habits & tools for reproducible research

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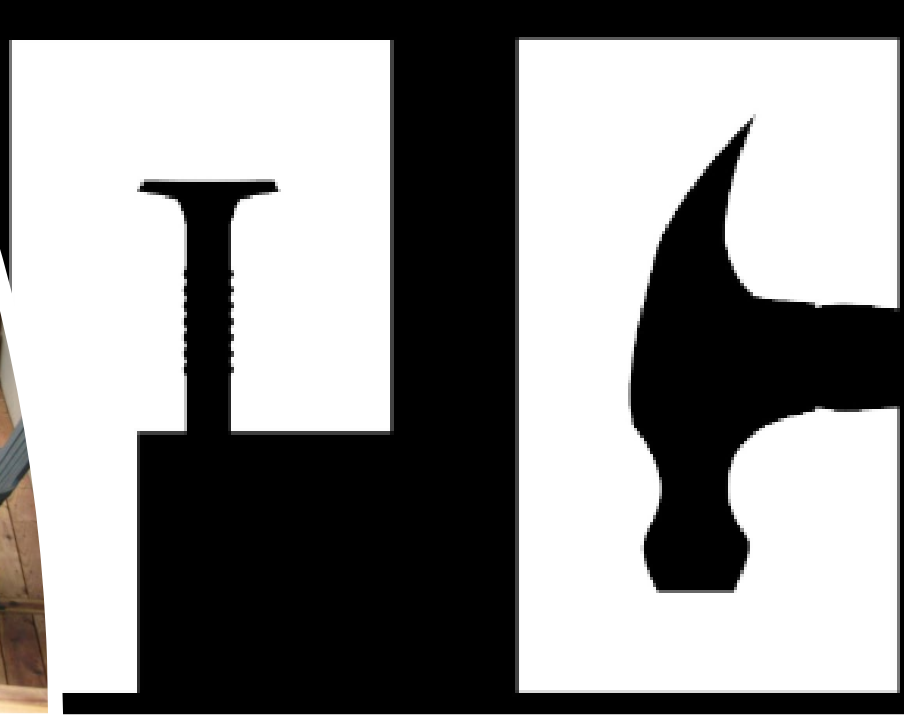
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Disclaimer: the views presented here are those of the speaker, and do not necessarily reflect the views of the EPA or the U.S. Government

A little about me

- I grew up building things, and I still do.







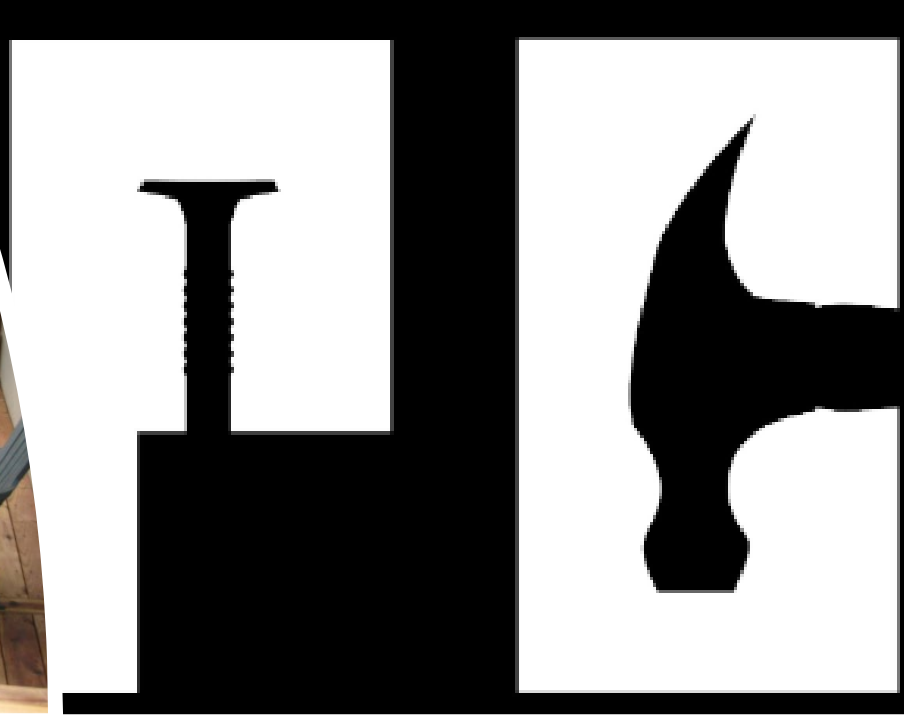


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- I grew up building things, and I still do.
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- I grew up building things, and I still do.
- My father's advice was drilled into me at an early age:

  "A clean workspace produces clean work."
- I don't build houses anymore, but I do build research!









Work Habits

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  - Do what works for you!
  - Recognize that this changes over time
  - Periodically reflect and update your approach

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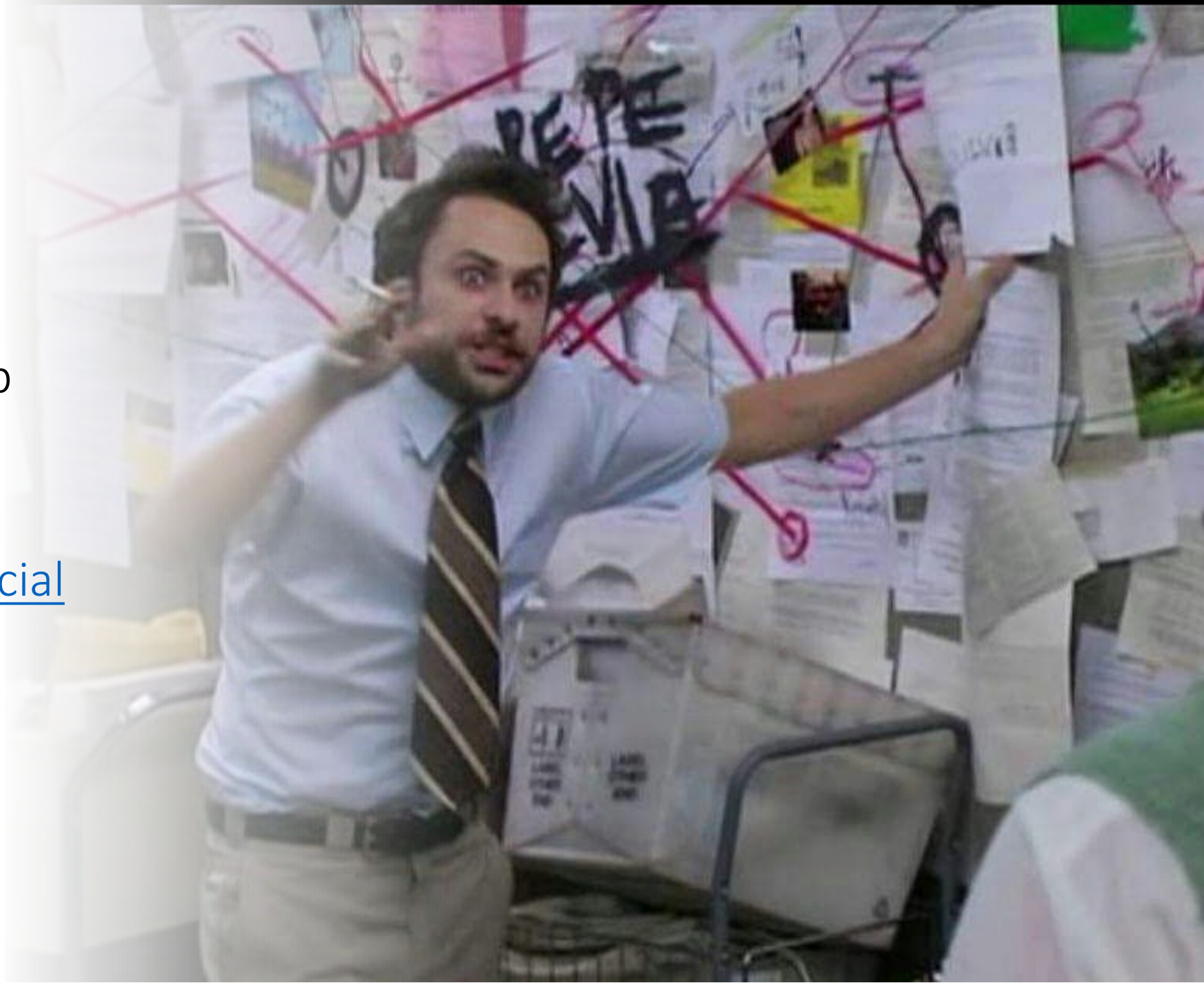
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- Things to consider:
  - Project priority (and life priorities)
  - Workflow
  - Switching costs

Directory Organization

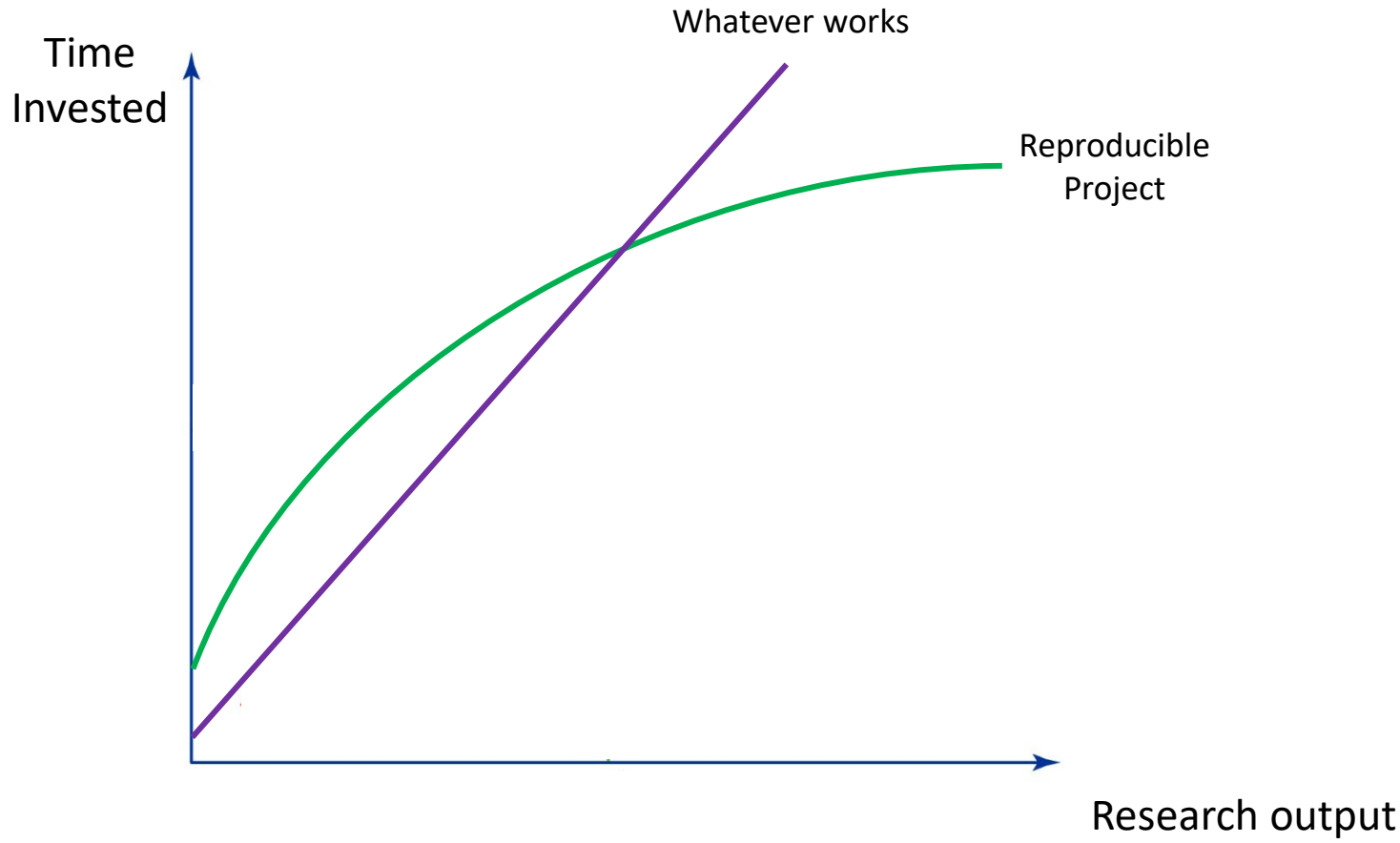
- Where do you keep your files?

Example with the Social Cost of Carbon



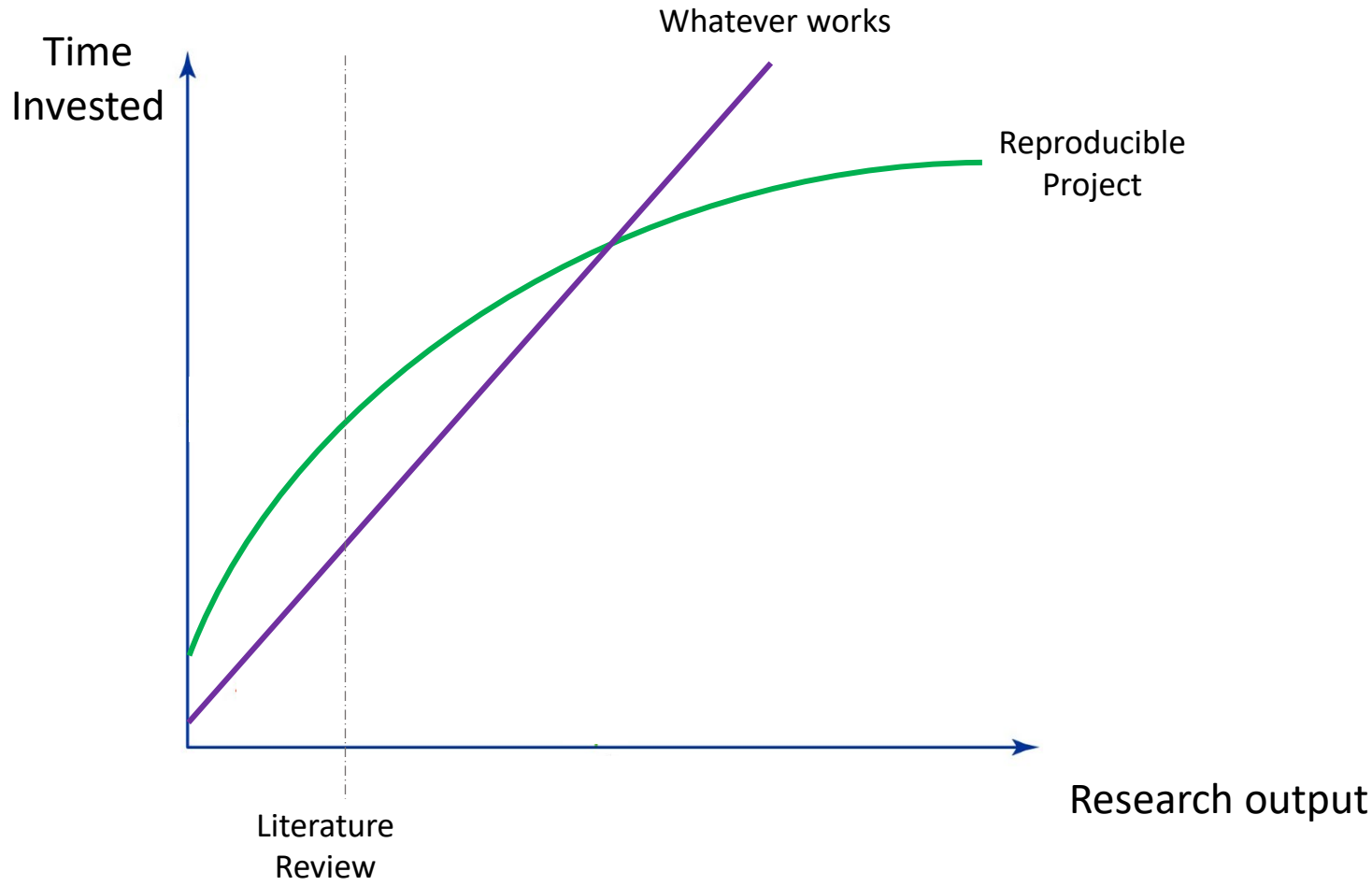
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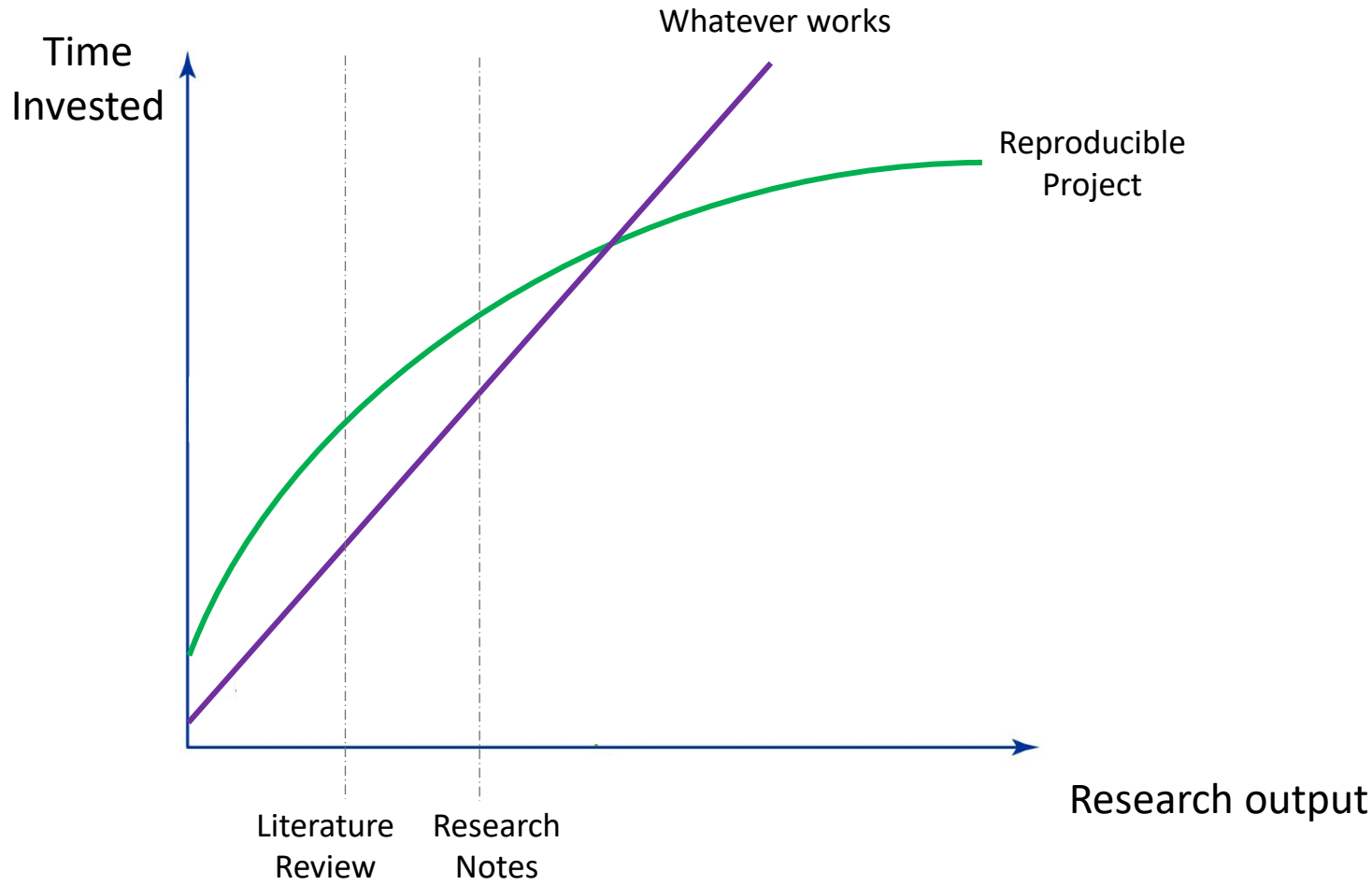
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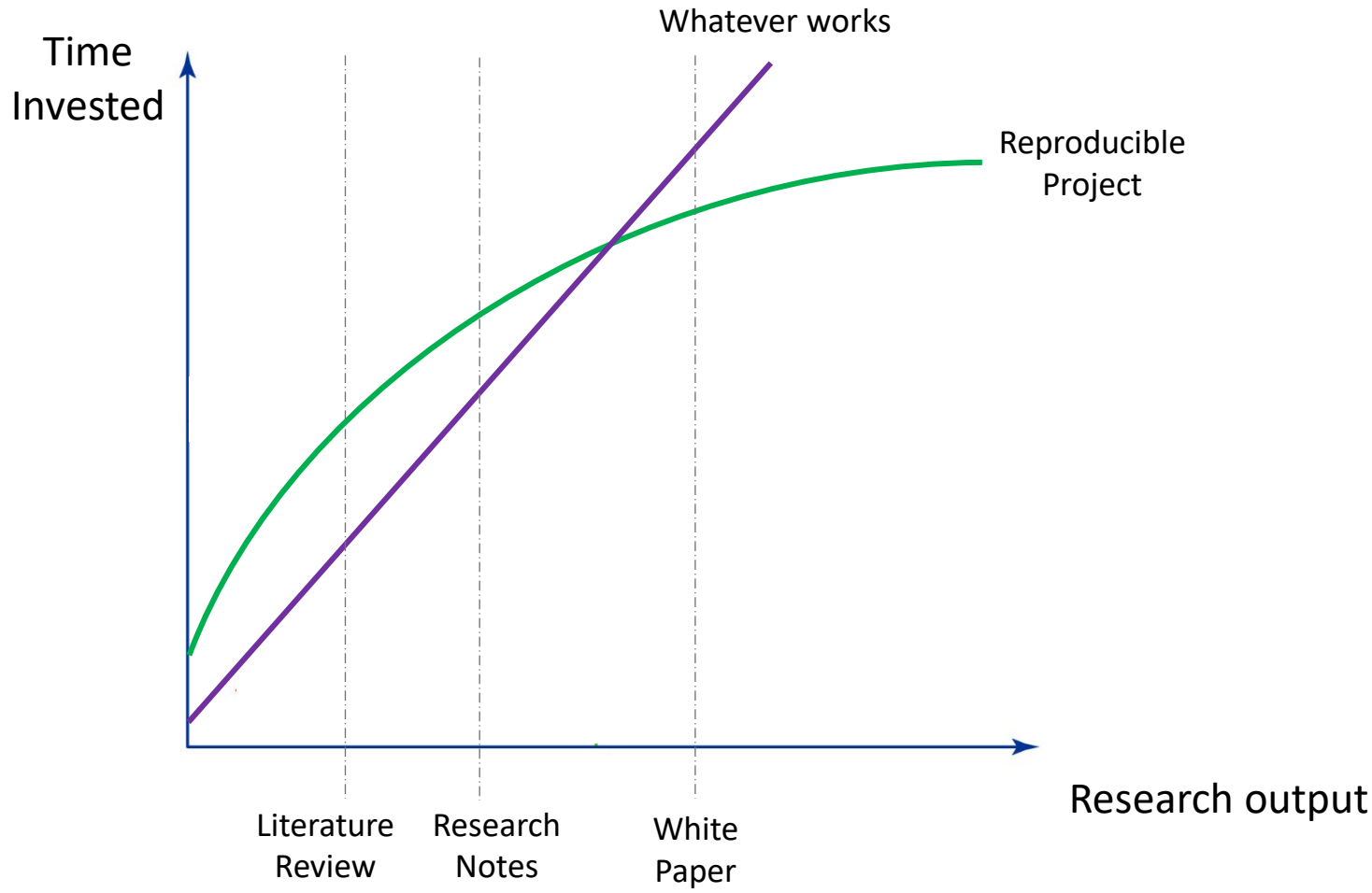
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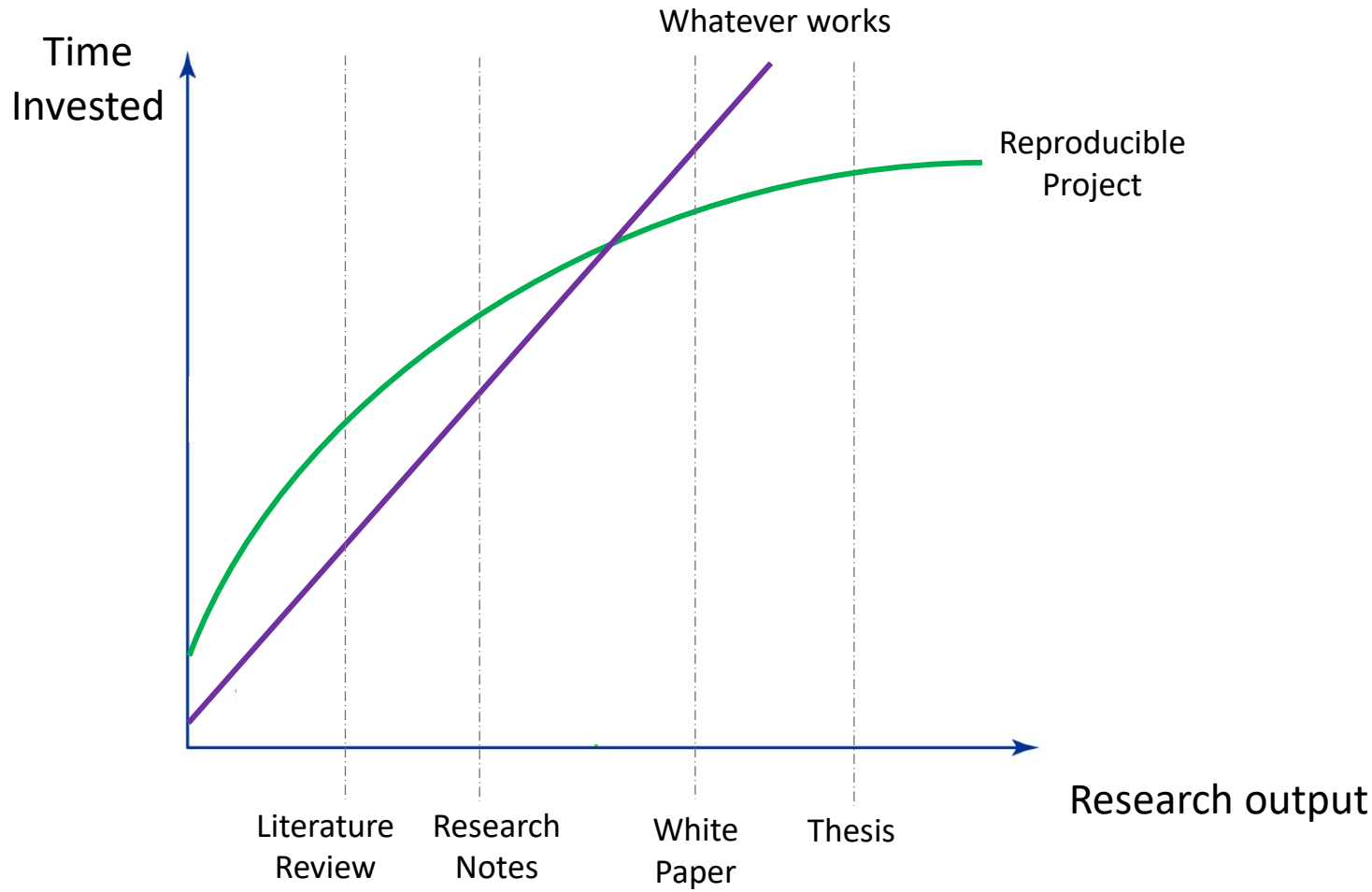
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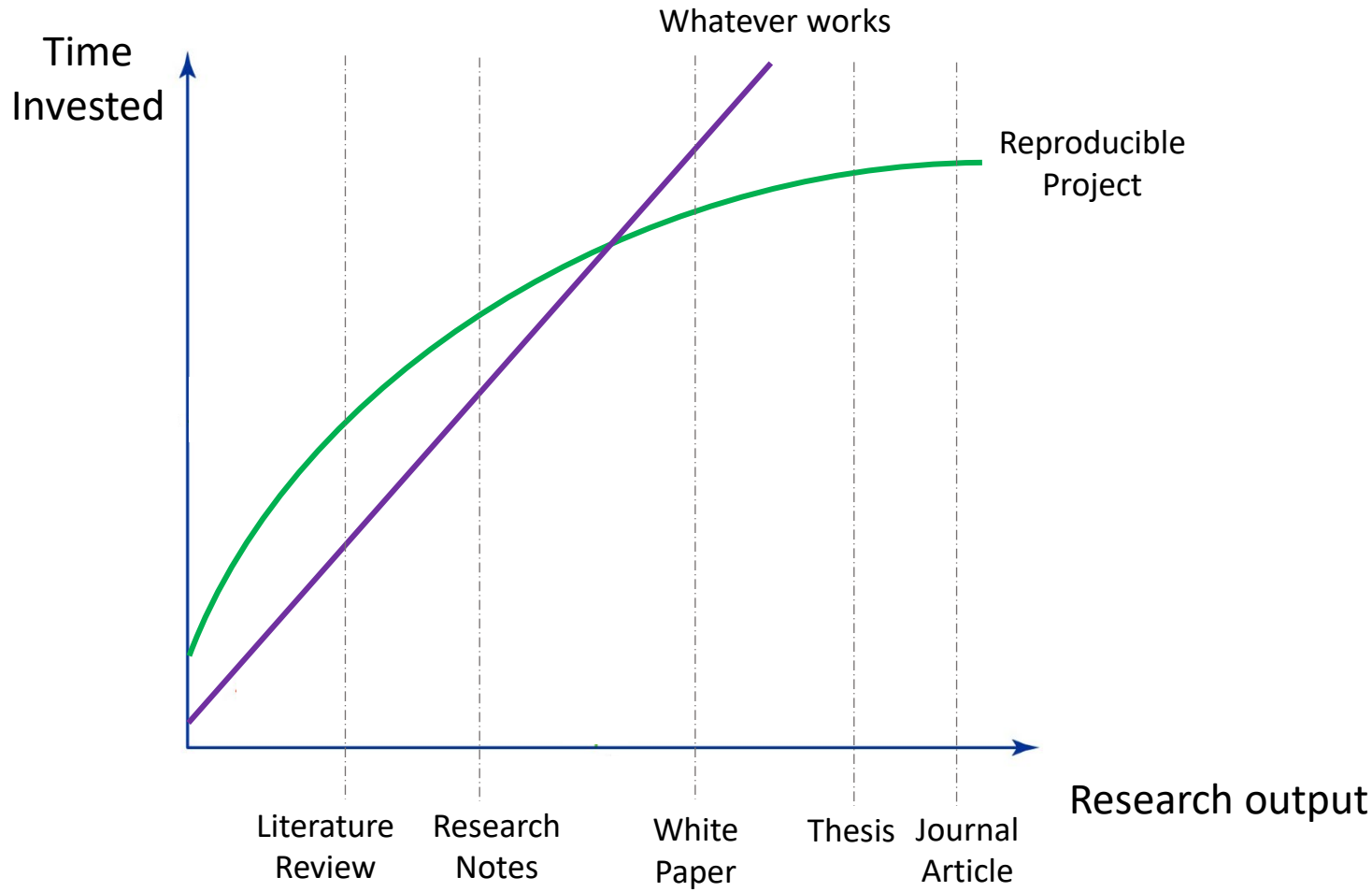
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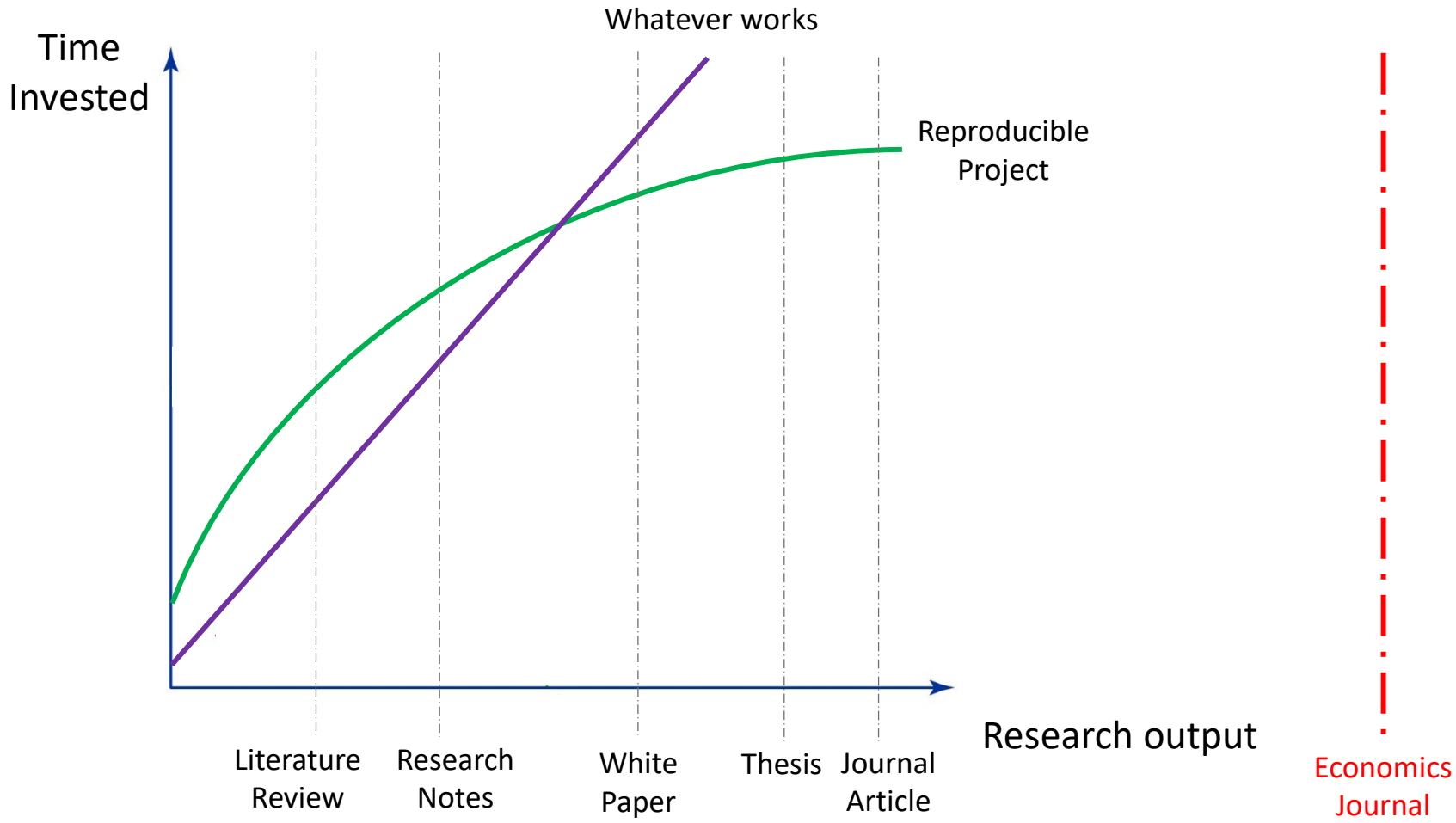
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- You have a research idea!!!
  - What are some things that you use to get started?
- Okay great, you want a research paper
  - Knowing this… what are your expectations?
    - Timeframe, workflow, process, etc.



Tools for Reproducible Research

"A clean workspace produces clean work"

- Integrated Development Environments (IDE's)
  - RStudio
    - The R programming language, although can be used with others
  - Visual Studio Code (VScode)
    - Many programming languages, typically Python, Julia, HTML, Java
  - Jupyter Notebooks
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  - Spyder
    - Python (but maybe others?)









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[Example problem set assignment]





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You really shouldn’t live without this



Example Repositories

The Social Cost of Greenhouse Gases

Hydrofluorocarbons and the Kigali Amendment

Willingness to ay to Improve Surface Water Quality

Greenhouse Gas Impact Value Estimator (GIVE)

PRISM Weather Data Extraction

North American Ski Resorts